

Chapter 6 — Univariate and bivariate visualization

Graphics reveal patterns tables hide

Learning objectives

- Choose univariate plots for distribution checks
- Read a correlation heatmap at a screening level

Why visualize

- Visualization supports pattern recognition, quality assessment, hypothesis generation
- In EDA, charts often detect missingness patterns, odd scales

Univariate plots

- Histograms show frequency by bin, useful for age or spend shape
- Box plots summarize median, quartiles, and outlier candidates via the IQR rule
- Pie charts work for quick share checks with few categories

Example 6.2 — Skewness informs model choice

- Example 6.2 — hands-on module
- Example 6.2 ties histogram or density shape to later methods
- Explore the chapter example module
- View files: ``modules/chapter6/example2/``

Bivariate plots

- Scatter plots expose linear or nonlinear association, clusters
- Correlation matrices store pairwise coefficients

Categorical–numerical views

- Box plots by category compare spend across product types
- These views support segment comparisons that group-by tables started numerically

Takeaways

- Match chart type to variable count and question
- Univariate plots check shape and outliers; bivariate plots screen relationships
- Correlation heatmaps prioritize pairs for deeper analysis in later parts

Next

- Complete the quiz for this part
- Continue to multivariate and high-dimensional visualization